

Problem 1

Show that the assumption that $\mathbb{P}$ is countably additive is equivalent to the assumption that $\mathbb{P}$ is continuous. That is to say, show that if a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfies $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, and $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ whenever $A, B \in \mathcal{F}$ and $A \cap B = \emptyset$, then $\mathbb{P}$ is countably additive (in the sense of satisfying Definition (1.3.1b)) if and only if $\mathbb{P}$ is continuous (in the sense of Lemma (1.3.5)).

Proof.

□

Problem 2

The 'ménages' problem poses the following question. Some consider it to be desirable that men and women alternate when seated at a circular table. If n heterosexual couples are seated randomly according to this rule, show that the probability that nobody sits next to his or her partner is

$$\frac{1}{n!} \sum_{k=0}^n (-1)^k \frac{2n}{2n-k} \binom{2n-k}{k} (n-k)!$$

You may find it useful to show first that the number of ways of selecting k non-overlapping pairs of adjacent seats is $\binom{2n-k}{k} 2n(2n-k)^{-1}$.

Proof.

□

Problem 3

The probabilistic method. 10 per cent of the surface of a sphere is coloured blue, the rest is red. Show that, irrespective of the manner in which the colours are distributed, it is possible to inscribe a cube in S with all its vertices red.

Proof.

□

Problem 4

Poker. During a game of poker, you are dealt a five-card hand at random. With the convention that aces may count high or low, show that:

$$\begin{aligned}
\mathbb{P}(\text{1 pair}) &\simeq 0.423, \\
\mathbb{P}(\text{straight}) &\simeq 0.0039, \\
\mathbb{P}(\text{4 of a kind}) &\simeq 0.00024, \\
\mathbb{P}(\text{2 pairs}) &\simeq 0.0475, \\
\mathbb{P}(\text{flush}) &\simeq 0.0020, \\
\mathbb{P}(\text{3 of a kind}) &\simeq 0.021, \\
\mathbb{P}(\text{full house}) &\simeq 0.0014, \\
\mathbb{P}(\text{straight flush}) &\simeq 0.000015.
\end{aligned}$$

Proof.

□

Problem 5

Let m be Lebesgue measure on $[0,1]$, and $0 \leq a \leq b \leq c \leq d \leq 1$ such that $a + d \geq b + c$. Give an example of a sequence of sets $A_1, A_2, \dots$ in $[0,1]$, such that $m(\liminf_n A_n) = a$, $\liminf_n m(A_n) = b$, $\limsup_n m(A_n) = c$ and $m(\limsup_n A_n) = d$.

Proof.

□

Problem 6

Let $\Omega = \mathbb{R}$ and consider the following subsets of $\mathcal{P}(\mathbb{R})$:

$$\begin{aligned}
\mathcal{C}_1 &:= \{(-\infty, b] : b \in \mathbb{R}\} \\
\mathcal{C}_2 &:= \{(a, b] : a, b \in \mathbb{R}\} \\
\mathcal{C}_3 &:= \{A \subset \mathbb{R}, A \text{ is closed}\}.
\end{aligned}$$

Show that $\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2) = \sigma(\mathcal{C}_3)$.

Proof.

□